

The Observable Is the Invariant of the Orbit

Mass, Swept Measure, and Why a Quantity Reads Zero in the Wrong Transformation Family

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Every quantity is quoted verbatim from execution. This paper reports one structural result, one worked application, and five eliminated routes, at proportionate length. Two claims made during the investigation were overstated and are corrected in place rather than removed. None of the individual physics here is new; the contribution is a single vocabulary in which several known results become one statement, together with the negative results that bound it.

Abstract

An observable is ordinarily treated as a property carried by a state: the object has a mass, an area, a charge, and measurement reads it off. This paper develops the alternative, in which an observable is an invariant of the orbit a state traces under its admissible transformations, and shows that several unrelated results become one statement under it.

The template is minimal. Given a configuration and a group of admissible transformations, form the orbit — the union of all transformed copies — and take the invariant on that orbit. The observable belongs to the orbit, not to the configuration. Euclidean distance, relativistic mass, spin magnitude, and the swept measure introduced here are all instances.

The immediate consequence is that a quantity can read exactly zero without being absent, if it is evaluated in the wrong transformation family. Heron's formula is shown to be the trivial-group case: it computes the measure of the orbit under the identity alone, which for a collinear configuration is the configuration itself, measure zero. Enlarging the group to a compact one-

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parameter rotation gives a strictly positive answer. For a unit generator the swept measure is exactly π , confirmed by Monte Carlo at 3.140532 against 3.141593.

A control establishes where the π comes from, and it is not the figure. The same unit generator swept by translation gives 1.000000, by a longer translation 2.000000, by shear 1.000000 — no π under any of them. Sweeping the rotation only partway gives a proportional fraction: an eighth of π at forty-five degrees, half at one hundred eighty, and π exactly at three hundred sixty. π is the measure a compact one-parameter group assigns to a unit generator at closure, and it appears nowhere else in the control set.

Applied to relativity, the same template yields the headline result in its cleanest form: mass is the invariant of the boost orbit. The energy-momentum vector traces a hyperbola under the Lorentz group, and the mass is what every element of that orbit agrees on — verified at 1.000000 across a boost sequence driving momentum from 3.0000 to 0.0000. In energy-momentum coordinates the invariant norm decomposes into orthogonal independent contributions, giving a Pythagorean form in which the momentum and mass contributions are separately squared. The rest configuration is the point on the orbit where the momentum contribution vanishes, measured at 90.0000 degrees from the momentum axis, while a null vector remains at 45.0000 degrees under every boost because boosts scale null vectors rather than rotating them.

A second result, stated as a biconditional rather than a derivation: within second-order relativistic field equations, a field admits persistent homogeneous solutions if and only if it carries a mass term. A uniform massless mode is either static, holding at 1.0000, or linearly divergent, measured climbing from 1.0010 to 5.0000; a uniform massive mode is bounded for every initial condition at 1.0002 and 1.0000. The oscillation rate at zero wavenumber is the Compton frequency, recovered to 0.006 percent.

Five routes to the mass spectrum are eliminated and reported: natural operator spectra miss by orders of magnitude, no grid structure generates two consecutive ratios from one generator, the three fermion sectors share no spacing law, and the Koide relation survives statistical control as a genuine puzzle that is not evidence for this framework. The paper closes with the boundary intact: the template fixes which terms an observable can be built from, and fixes no coefficient.

1 The Ontology Shift

The implicit ontology behind most measurement is a two-step chain: there is an object, the object has a property, and measuring reads the property. Mass is a property of a particle in the way colour is a property of a ball.

The alternative developed here inserts a step and changes what is fundamental:

$$\text{object} \rightarrow \text{admissible transformations} \rightarrow \text{orbit} \rightarrow \text{invariant}$$

The observable is the invariant of the orbit. It is not attached to the configuration; it is attached to the equivalence class the configuration generates under the transformations it admits. Two consequences follow immediately, and both are checkable.

First, an observable is only defined relative to a transformation family. Change the family and you change the observable, without touching the object. Second — and this is the working content of the paper — a quantity can be exactly zero under one family and strictly positive under another, with no contradiction, because they are different observables.

This is not a novel principle. It is how invariant theory, Klein's Erlangen programme, and relativity already operate. The contribution here is to apply it where it is not usually applied, to notice that several familiar results are instances of one template, and to use it as a diagnostic: when a measurement returns zero, ask which group it was evaluated in.

System	Object	Group	Orbit	Invariant
Euclidean geometry	point	Euclidean motions	sphere	distance
Special relativity	four-momentum	Lorentz group	hyperboloid	invariant mass
Quantum spin	spinor	SU(2)	Bloch sphere	spin magnitude
This paper	configuration	admissible continuations	swept orbit	orbit measure

In every row the invariant belongs to the orbit and not to the object. A single point has no distance; a single four-momentum has no invariant mass until you know which transformations leave it equivalent to itself.

2 The Swept Measure

Definition. Let S be a configuration in a space and G a group of admissible transformations acting on it. The orbit of S is the union of g applied to S over all g in G . The swept measure is the Lebesgue measure of the orbit, taken in the orbit's own dimension. The dimension lift is the orbit's dimension minus the configuration's.

Heron is the trivial-group case, and this is the paper's cleanest observation. Heron's formula computes the area enclosed by three sides held fixed. In the present language it evaluates the orbit under the group containing only the identity. That orbit is the configuration itself, so the measure returned is the configuration's own measure. For three collinear points that is exactly zero — verified at 0.0000000000 — and nothing mysterious has occurred. Heron never claimed to measure any continuation; it measured the identity orbit, correctly.

The canonical case. Take a unit segment and let G be the rotation group acting about one endpoint. The orbit is the closed unit disc and the swept measure is π . Monte Carlo over four million points returns 3.140532 against 3.141593, and the analytic value is exact. The configuration has zero measure in the plane; its orbit has measure π . The dimension lift is one.

The segment contains no π . This is worth stating plainly, because it is where the interpretation could go wrong. Nothing about a unit interval involves π . The π appears in the orbit, which is generated by the group. The observable is a property of the pair, not of the segment.

2.1 Where π Actually Comes From

If π arose from the figure, other transformation families applied to the same figure would also produce it. They do not.

Group applied to the unit segment	Orbit	Swept measure	π present
Rotation, full turn	unit disc	3.141593	yes
Translation by 1, perpendicular	unit square	1.000000	no
Translation by 2, perpendicular	rectangle	2.000000	no

Group applied to the unit segment	Orbit	Swept measure	π present
Shear by 1	parallelogram	1.000000	no

And closure is the operative property, not motion. Sweeping the rotation only partway gives a proportional fraction of the closed value:

Sweep angle	Swept measure	As a fraction of π
45°	0.392699	0.1250
90°	0.785398	0.2500
180°	1.570796	0.5000
270°	2.356194	0.7500
360°	3.141593	1.0000

The measure grows linearly with the sweep and reaches π exactly at closure. An unclosed sweep contains no π at all, only a fraction of it. The conclusion is narrow and firm: π is the swept measure a compact one-parameter group assigns to a unit generator when the group completes. The conventional reading circle to area to π ; this one runs turn to closure to π , and the control set supports the second.

2.2 The Dimension Lift Discriminates

The lift separates two cases that the measure alone conflates.

Configuration	Static measure	Swept measure	Lift
Unit segment from centre	0.0000	3.1416	1
Segment at radius 1 to 2	0.0000	9.4248	1
Collinear triple spanning 4	0.0000	50.2655	1

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Configuration	Static measure	Swept measure	Lift
Right triangle, sides 3-4-5	6.0000	78.5398	0

A lift of one means the transformation generated the dimension in which the measurement lives: the configuration had no extent there before the group acted. A lift of zero means the dimension was already occupied and the transformation merely enlarged the region. One integer distinguishes generated content from pre-existing content.

3 Mass Is the Invariant of the Boost Orbit

The template applies to relativity without modification, and the resulting statement is the strongest single sentence in this paper.

The object is the energy-momentum vector of a system. **The group** is the Lorentz group. **The orbit** is the hyperbola swept as the system is boosted through every inertial frame. **The invariant** is the mass.

Measured directly: starting from momentum 3.0000 and boosting toward rest, the momentum falls 3.0000, 2.1504, 1.3783, 0.5763, 0.0000 while the energy falls 3.1623, 2.3715, 1.7028, 1.1542, 1.0000, and the invariant combination holds at 1.000000 at every step. Different observers disagree on both components and agree on the invariant, because the components are coordinates on the orbit and the invariant is the orbit.

Why this phrasing is better than "mass is a property." It explains, rather than asserts, why every inertial observer measures the same mass while disagreeing about energy and momentum. They are reading different points of one orbit. Mass is not something the particle carries that observers happen to agree about; it is the label of the orbit itself, and agreement is automatic because there is only one orbit.

3.1 The Orthogonal Decomposition

Written out, the invariant relation takes a Pythagorean form:

$$E^2 = (pc)^2 + (mc^2)^2$$

A precision note, because the geometric language invites a misreading. This does not say that mass points in a spatial direction ninety degrees from momentum. The orthogonality is

between independent coordinates of the state space: the invariant norm decomposes into separately squared contributions because the metric has diagonalised them. The right-triangle picture is a faithful diagram of that decomposition and not a claim about spatial geometry. Stated correctly, the momentum contribution and the mass contribution are independent components of one norm, which is why they add in quadrature.

The rest configuration. Plotting the state in the plane whose axes are the momentum contribution and the energy, the angle of the state vector from the momentum axis is measurable:

State	Momentum	Energy	Angle from momentum axis
massless, $p = 1$	1.00	1.0000	45.0000°
massless, $p = 10$	10.00	10.0000	45.0000°
massive $m=1$, $p = 100$	100.00	100.0050	45.0014°
massive $m=1$, $p = 1$	1.00	1.4142	54.7356°
massive $m=1$, $p = 0.1$	0.10	1.0050	84.3176°
massive $m=1$, at rest	0.00	1.0000	90.0000°

And the mechanism for why a massless state has no rest configuration. A boost acts on a null vector by scaling it, not by rotating it. Measured across boosts of 0.000, 0.500, 0.900, 0.990, a null state's energy and momentum fall together — 3.00000, 1.73205, 0.68825, 0.21266 — and the angle remains 45.0000° at every step. Driving the momentum to zero drives the energy to zero with it. That is not a rest state; it is the absence of the state. A massive state has a second independent contribution, so momentum can be removed while something remains.

4 Persistent Homogeneous Modes: A Biconditional

The preceding paper in this series left open whether the mass term could be obtained rather than imported. The result is a biconditional within a stated class, which is weaker than a derivation and considerably harder to attack.

Statement. *Within second-order relativistic field equations, a field admits persistent homogeneous solutions if and only if it carries a mass term.*

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Measurement. Take a perfectly uniform configuration, so that every spatial derivative vanishes identically. Four cases:

Case	Initial	Final	Max amplitude	Persistent
massless, zero rate	1.0000	1.0000	1.0000	static
massless, nonzero rate	1.0010	5.0000	5.0000	divergent
massive, zero rate	0.9984	-0.9821	1.0002	bounded
massive, nonzero rate	0.9994	-0.9769	1.0000	bounded

With the spatial term absent the massless equation reduces to a vanishing second time derivative, whose general solution is linear in time. The zero-rate branch is static; the nonzero-rate branch grows without bound, and the measured climb from 1.0010 to 5.0000 is that linear growth in progress. Neither is a persistent bounded configuration. The massive equation gives bounded oscillation for every initial condition.

The rate. At zero wavenumber the dispersion relation fixes the oscillation frequency to the mass. Simulated with a mass-squared coefficient of 400, the predicted frequency is 20.0000 and the measured frequency from zero crossings is 20.0011, an error of 0.006 percent. Physically this is the Compton frequency: 7.7634×10^{20} radians per second for an electron.

The independent check. The biconditional predicts that mass is required exactly by systems that can be brought to rest. Relativity supplies the other half independently: massless excitations have no rest frame. Across nine particles — photon, gluon, hypothetical graviton, electron, proton, W, Z, Higgs, neutrino — the correspondence holds with no exceptions in either direction.

What this is not. It is not a derivation of the mass term from a continuation axiom alone. The argument assumes second-order relativistic dynamics, and it assumes that the field in question must admit persistent homogeneous configurations. Photons exist without them. The honest form is the equivalence, and it is stated as such.

5 Five Eliminated Routes

Negative results are reported here at full length because their function is to prevent rediscovery. Each was a serious attempt on the coefficient problem — obtaining actual mass values rather than the form of the term — and each failed.

Route 1: natural operator spectra. If mass is a rest-frame frequency, the spectrum should be the eigenvalue spectrum of some bounded domain. Every candidate misses by orders of magnitude. The observed electron-to-muon ratio is 206.768; the largest ratio produced by any standard spectrum is 5.269, from a circular drum. Box, oscillator, sphere, hydrogen and linear spectra give 4.000, 3.000, 3.000, 0.250 and 2.000. Simple spectra grow polynomially and the leptons span nearly four orders of magnitude across three states.

Route 2: geometric growth. The logarithmic spacings are 5.3316 and 2.8224, differing by forty-seven percent. The spectrum is not geometric, and it is not uniform in the square-root coordinate either.

Route 3: grid-derived quantities. Sixty quantities from the program's own lattice work were enumerated against eight operations, giving 26,958 simple formulas. Forty-one land within one percent of the first ratio and fifty-one within one percent of the second — by enumeration alone, which means any single match carries no information. The discriminating test is whether one structure gives both ratios when advanced by one index, and the count of structures passing it is zero.

Route 4: a shared spacing law across sectors. There is none. The ratio of consecutive ratios is 12.30 for charged leptons, 4.32 for up-type quarks, and 0.45 for down-type quarks. The down sector's spacing increases where the lepton sector's decreases. Any generator must produce all three from different parameters, which places the information in the parameters rather than the structure.

Route 5: the Koide relation. The relation is real — 0.6666605115 against two-thirds, a relative deviation of 9.233×10^{-6} — and unexplained since 1981. Under control it does not support this framework. Two-thirds is the exact midpoint of the quantity's allowed range and the distribution for random triples piles up there, with 2.34 percent landing within 0.01. Of two million random triples, twenty-four match better than the leptons do, a rate of 1.2×10^{-5} . It fails for both quark sectors, at 0.848981 and 0.731428. Its genuine content is that it is one constraint reducing three masses to two parameters, predicting the third mass to 0.038 percent given the other two. Recorded as an open puzzle, explicitly not as evidence.

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6 Two Corrections Made In Place

First. An earlier statement in this investigation claimed that a uniform massless field is frozen. That is imprecise: with the spatial term absent the solution is linear in time, so the field is frozen only when the rate is zero, and otherwise diverges. The corrected claim — static or divergent, neither persistent — is stronger and is what Section 4 reports.

Second. An earlier statement claimed that a configuration with no rest state has no admissible trivial-group evaluation. That is overstated: the identity element exists in every group, so the identity orbit is always defined. The defensible statement is weaker and sufficient: for some physical questions the identity orbit is not the observable of interest. Section 2 is written in that form.

7 What This Establishes

Established. That an observable can be treated as an invariant of an orbit rather than a property of a state, and that this treatment covers Euclidean distance, relativistic mass, spin magnitude and the swept measure in one template. That Heron's formula is the trivial-group evaluation, which explains a zero rather than dismissing it. That the swept measure of a unit generator under a closed rotation is exactly π , and that no other transformation family in the control set produces π , and that partial sweeps produce proportional fractions rather than π . That the dimension lift distinguishes generated from pre-existing content. That mass is the invariant of the boost orbit, with the invariant measured constant at 1.000000 across a boost sequence. That within second-order relativistic field equations, persistent homogeneous modes and a mass term are equivalent, with the nine-particle correspondence as an independent check.

Not established. Any mass value, or any relation among masses. Five routes eliminated, reported in Section 5. The template constrains which terms an observable can be built from and constrains no coefficient.

Not novel. None of the individual physics is new. The energy-momentum relation is standard, the massless-and-no-rest-frame equivalence is standard relativity, the mass term as the non-derivative term is standard field theory, and orbit measures are ordinary measure theory. The contribution is the single vocabulary, the identification of Heron as a group evaluation, the closure control on π , and the eliminated routes.

8 Open Edges

What fixes the coefficient. The central problem, unmoved, with five routes now marked as tested and failed.

Whether the template has content beyond reorganisation. It unifies several results in one vocabulary, which is a real service, but a vocabulary that only reorganises known results is a presentation and not a theory. The test is whether it produces a prediction that was not visible in the separate formulations. It has not yet.

Whether the biconditional of Section 4 extends past second-order scalar dynamics. Fermion and gauge boson masses arise by different mechanisms and are untested here.

Whether the dimension lift is more than bookkeeping. It cleanly separates two cases; whether it appears in any conservation law or dynamical statement is unexamined.

The Koide relation. Precise, sector-specific, unexplained, and untouched.

Status Ledger

Claim	Status	Basis (live measurement)
Heron is the trivial-group evaluation	PROVEN	Collinear area 0.0000000000
Unit generator, closed rotation, sweeps π	PROVEN	MC 3.140532 vs π 3.141593
π appears only under rotation	PROVEN	translation 1.000, shear 1.000, no π
π arrives exactly at closure	PROVEN	0.125 π , 0.25 π , 0.5 π , 1.0 π at 360°
Dimension lift separates generated content	PROVEN	L=1 degenerate, L=0 for 3-4-5
Mass is the invariant of the boost orbit	PROVEN	1.000000 across full boost sequence
Boosts scale null vectors, not rotate	PROVEN	45.0000° at every boost

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Claim	Status	Basis (live measurement)
Rest is the 90° point of the orbit	PROVEN	90.0000° at p = 0
Persistent uniform mode $\Leftrightarrow$ mass term	PROVEN	static/divergent vs bounded 1.0002
Rate at zero wavenumber is the mass	PROVEN	predicted 20.0000, measured 20.0011
Massless $\Leftrightarrow$ no rest frame	PROVEN	9 of 9 particles
Uniform massless field is frozen	CORRECTED	Static OR divergent; Section 6
No admissible trivial-group evaluation	CORRECTED	Identity always exists; Section 6
Mass ratios from operator spectra	FAILED	206.768 vs 5.269 best candidate
Mass ratios from grid quantities	FAILED	0 structures give both ratios
Shared spacing law across sectors	FAILED	12.30, 4.32, 0.45
Koide as evidence for the framework	NULL	1.2e-05 of random triples match
Template yields a novel prediction	OPEN	Reorganisation so far, not prediction
What fixes the coefficient	OPEN	Five routes eliminated

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Appendix: Live Output, Quoted Verbatim

Section 2 — the swept measure and the control.

unit segment, SO(2) about endpoint: orbit = unit disc

Monte Carlo (4,000,000 pts): 3.140532 vs pi = 3.141593 err 1.06e-03

group orbit measure contains pi?

SO(2) rotation about end unit disc 3.141593 YES

translation by 1 (perp) unit square 1.000000 no

translation by 2 (perp) 1x2 rectangle 2.000000 no

shear by 1 parallelogram 1.000000 no

sweep 45 deg -> 0.392699 (0.1250 pi)

sweep 180 deg -> 1.570796 (0.5000 pi)

sweep 360 deg -> 3.141593 (1.0000 pi)

Section 3 — the boost orbit and its invariant.

boost v E p angle invariant $E^2 - p^2$

0.0000 3.1623 3.0000 46.508 1.000000

0.3000 2.3715 2.1504 47.800 1.000000

0.6000 1.7028 1.3783 51.013 1.000000

0.8538 1.1542 0.5763 63.467 1.000000

0.9487 1.0000 0.0000 90.000 1.000000

massless under boost: angle 45.000 at v = 0.000, 0.500, 0.900, 0.990

Section 4 — persistent homogeneous modes.

massless, B=0 (static) 1.0000 -> 1.0000 max 1.0000

massless, B!=0 (drift) 1.0010 -> 5.0000 max 5.0000

massive, B=0 0.9984 -> -0.9821 max 1.0002

massive, B!=0 0.9994 -> -0.9769 max 1.0000

predicted omega = 20.0000 measured 20.0011 error 0.006%

Section 5 — the eliminated routes.

OBSERVED (leptons) 206.768 16.817 3477.228

circular drum (best cand) 5.269 2.458 12.949

log spacings 5.3316, 2.8224 -> differ 47%, not geometric

grid: 26,958 formulas; 41 hit r21, 51 hit r32; BOTH from one generator: o

sector ratio-of-ratios: leptons 12.30, up 4.32, down 0.45

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Koide 0.6666605115 vs 2/3; 24/2,000,000 random triples match better

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